

Article

Optimizing Energy Efficiency in Last-Mile Delivery: A Collaborative Approach with Public Transportation System and Drones

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Abstract

Accurately estimating the energy consumption of unmanned aerial vehicles (UAVs) in real-world delivery scenarios remains a critical challenge, particularly when UAVs operate in complex urban environments and are coupled with public transportation systems. Most existing models rely on oversimplified assumptions or static mission profiles, limiting their applicability to realistic, scalable drone-based logistics. In this paper, we propose a physically-grounded and scenario-aware energy sizing methodology for UAVs operating as part of a last-mile delivery system integrated with a city's bus network. The model incorporates detailed physical dynamics—including lift, drag, thrust, and payload variations—and considers real-time mission constraints such as delivery execution windows and infrastructure interactions. To enhance the realism of the energy estimation, we integrate computational fluid dynamics (CFD) simulations that quantify the impact of surrounding structures and moving buses on UAV thrust efficiency. Four mission scenarios of increasing complexity are defined to evaluate the effects of delivery delays, obstacle-induced aerodynamic perturbations, and early return strategies on energy consumption. The methodology is applied to a real-world transport network in Belfort, France, using a graph-based digital twin. Results show that environmental and operational constraints can lead to up to 16% additional energy consumption compared to idealized mission models. The proposed framework provides a robust foundation for UAV battery sizing, mission planning, and sustainable integration of aerial delivery into multimodal urban transport systems.

Keywords: UAV energy modeling; last-mile delivery; public transport; CFD; battery sizing; urban logistics; aerial mobility; multimodal transport

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1. Introduction

The exponential growth of e-commerce and the rise of on-demand services have placed increasing pressure on urban logistics systems, particularly in the last-mile segment [1,2]. Traditional delivery vehicles contribute significantly to road congestion, noise, and environmental pollution [3,4], prompting a growing interest in alternative delivery solutions. In this context, unmanned aerial vehicles (UAVs) have emerged as a promising approach to reduce delivery times, bypass road traffic, and lower the carbon footprint of parcel distribution in dense urban areas [5,6].

Despite their advantages, UAV-based delivery systems face major challenges, especially regarding their limited energy autonomy, which restricts their delivery range and operational flexibility [7,8]. In response, hybrid approaches have been explored that couple

UAVs with ground vehicles to extend their effective range. A particularly promising concept involves integrating drones with public transportation systems (PTSs) such as buses or trains [9,10] enabling drones to “ride” on these vehicles and deploy near delivery locations, thereby reducing the need for long-range flights and eliminating frequent returns to depots.

However, existing works on UAV–PTS integration tend to focus primarily on routing strategies or high-level delivery schemes [10–12] while neglecting the accurate modeling of energy consumption. Most rely on idealized assumptions such as constant power draw, negligible drag forces, or perfect flight trajectories [8,13]. These simplifications lead to significant errors in battery sizing and mission feasibility assessment, especially in dynamic and constrained urban environments.

This leads to the core scientific challenge addressed in this paper: “How can we accurately predict the energy consumption of UAVs operating in realistic multimodal delivery systems, accounting for physical constraints and environmental disturbances, to enable robust battery sizing and mission planning?” [7].

To address this gap, we propose a physically-grounded, scenario-aware energy modeling framework tailored for UAVs operating in conjunction with public transport networks [12,14]. Our model combines Newtonian dynamics, aerodynamic forces, and mission constraints with computational fluid dynamics (CFD) simulations to correct for the influence of surrounding structures and moving vehicles on drone thrust [7,15]. We define four delivery scenarios of increasing complexity, including ideal flight, delivery time pressure, thrust disturbances due to urban obstacles, and early return maneuvers to synchronize with PTS schedules.

Our method is applied to a real-world case study using a graph-based digital twin of the bus network of Belfort, France, with UAV and bus data provided by the public transport operator. Results show that failing to consider delivery constraints and aerodynamic disturbances can lead to energy underestimation of up to 16%, which is critical for energy-aware mission planning and fleet reliability.

The contributions of this work are as follows:

- We develop a complete and transferable energy estimation methodology for UAV-based parcel delivery systems integrated with public transport networks.
- We integrate CFD-derived corrections into the physical model to account for real-world aerodynamic effects.
- We define and analyze four mission scenarios, each reflecting common operational constraints in urban delivery.
- We validate the approach on a real urban network, showing its applicability for battery sizing and sustainable aerial delivery planning.

The rest of this paper is organized as follows. Section 2 reviews the related literature on last-mile delivery, UAV–vehicle integration, and energy consumption modeling. Section 3 describes the system architecture and modeling assumptions. Section 4 presents the energy estimation framework, including dynamic modeling and CFD integration. Section 5 introduces the experimental setup and simulation parameters. Section 4 discusses the results and energy trends under different scenarios. Section 5 concludes and outlines directions for future research.

2. Related Work

2.1. Last-Mile Delivery and Urban Logistics

Last-mile delivery (LMD) is one of the most studied challenges in urban logistics due to its high operational cost, significant contribution to urban congestion, and environmental impact [6,9]. It represents the final segment of the supply chain, linking urban distribution centers (UDCs) to customers. In densely populated areas, the complexity of traffic condi-

tions, regulatory constraints, and customer expectations for rapid, reliable delivery make LMD both critical and costly.

To reduce these impacts, cities and logistics providers have experimented with alternative modes and strategies, such as relay points, electric vehicles, autonomous ground robots, and micro-hubs. Among these, unmanned aerial vehicles (UAVs) have emerged as a promising solution due to their ability to bypass road congestion and directly reach delivery points, particularly in difficult or high-density areas [5,8]. However, UAV deployment at scale is still limited by regulatory, safety, and energy constraints.

2.2. UAV-Based Delivery Systems and Integration with Transport Networks

Drone-based delivery systems have evolved from prototype experimentation to structured deployments led by both industrial actors and academic researchers. Several companies have demonstrated UAV use for short-range logistics, and studies have proposed various delivery architectures addressing routing, task assignment, and drone–vehicle coordination [7,14].

A particularly promising approach involves hybrid delivery architectures, where UAVs are paired with terrestrial vehicles, such as trucks or vans, to increase delivery coverage and optimize fleet efficiency. In these models—commonly formalized as variants of the traveling salesman problem with drones (TSPD) or vehicle routing problems (VRPs)—the drone is used for short detours from the main vehicle’s path [5,14].

An alternative line of work investigates the integration of UAVs with public transport systems (PTSs), such as buses or trains. In these systems, drones “ride” on public vehicles and deploy near the delivery locations. Several studies have modeled this concept using deterministic or time-dependent routing approaches, and some have considered mission constraints such as delivery time windows or limited battery capacity [10,11]. Other works explore the use of PTS as mobile recharging platforms, enabling drones to extend their autonomy through scheduled energy replenishment [16,17].

These contributions demonstrate the potential of UAV–PTS synergy, but their focus remains primarily on feasibility and high-level routing logic, often abstracting away the physical and energetic realism of drone behavior during deployment.

2.3. Energy Consumption Models for UAVs

Energy consumption modeling is a critical component in UAV mission planning and system design. Many existing approaches rely on simplified assumptions such as constant cruising speed, negligible aerodynamic drag, or uniform energy consumption across phases of flight. These simplifications may be suitable for preliminary analysis, but they fail to capture the variability of real missions, especially in dense urban areas [7].

More physically informed models have emerged to describe UAV energy consumption based on thrust, drag, payload mass, and trajectory dynamics. These models use first-principle mechanics and include considerations for battery discharge rates, motor efficiency, and energy losses during takeoff and landing [7]. The sensitivity of UAV energy consumption to flight altitude, obstacle proximity, and urban wind conditions has been demonstrated in various simulation studies.

To further enhance realism, computational fluid dynamics (CFD) [15] has been used to model airflow interactions between UAVs and their environments. CFD enables fine-grained analysis of thrust variation caused by aerodynamic perturbations near buildings or moving vehicles. However, despite its analytical strength, CFD remains largely absent from mission-level energy modeling frameworks, particularly in the context of urban multimodal delivery systems involving PTS coupling.

This observation motivates the present work, which introduces a scenario-aware energy modeling framework that combines physical modeling, environmental corrections, and mission-level constraints to support realistic energy estimation and system planning for UAV-based delivery integrated with public transport infrastructure.

3. Modeling Framework

This section presents the modeling framework developed to estimate the energy consumption of a UAV-based delivery system integrated with a public transport network. We first describe the proposed delivery architecture and operational assumptions. Then, we introduce the physical energy model based on the Fundamental Principle of Dynamics (FPD), followed by the definition of four mission scenarios. Finally, we detail the aerodynamic correction method based on computational fluid dynamics (CFD) analysis.

3.1. System Description and Operational Assumptions

The proposed delivery architecture is based on a multimodal system that couples UAVs with a public transportation network, specifically the bus system of the city of Belfort, France. This system, illustrated in Figure 1, enables drones to perform last-mile deliveries by leveraging the existing circulation of public buses, thereby avoiding the need for dedicated delivery vehicles.

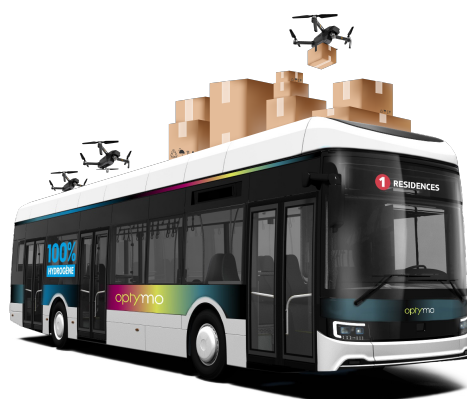


Figure 1. System architecture of the Deliv'air project: UAVs perform deliveries from the roof of public buses and rejoin the line for sequential missions.

Each drone is initially loaded onto the roof of a bus at a central logistics hub located near the train station, which acts as a depot and sorting point. Parcels are pre-loaded on the drone prior to departure. As the bus moves along its line, the drone monitors its GPS position. When approaching a delivery point (i.e., a bus stop with an associated delivery), the drone autonomously takes off from the bus roof, ascends vertically, performs a short horizontal flight to the customer's location (within a 15 m radius from the stop), and completes the delivery by lowering the parcel and notifying the recipient.

After delivery, the drone climbs to a cruising altitude, navigates horizontally to intercept the same or a following bus on the same line, and lands back on its roof. If necessary, the drone may change carriers (i.e., switch to another bus on the same line) to maintain operational continuity. Wireless recharging through inductive systems located on the bus roof is also envisioned to extend mission range and reduce dependency on full battery swaps.

The bus lines operate on a fixed schedule, with a bus dispatched every 10 to 15 min depending on the line. This dense cadence enables sustained parcel throughput without introducing new traffic on the urban road network.

The key assumptions of the system are summarized below:

- Deliveries are made sequentially at bus stops along a single line.

- The drone performs short point-to-point deliveries within a fixed radius from the bus stop.
- UAV takeoff and landing occur vertically from/to the moving bus roof.
- The drone does not return to a depot; it remains in operation along the bus line.
- Each drone mission includes takeoff, ascent, cruise, delivery, return cruise, and landing.
- Drone speed is assumed constant during horizontal segments; vertical ascent/descent durations are computed.
- No wind or weather effects are modeled at this stage.
- Payload mass varies from 1 to 10 kg depending on the scenario.

This architecture allows for continuous and energy-efficient delivery in urban areas while reducing the number of ground vehicles required and minimizing road congestion.

3.2. Energy Modeling Based on the Fundamental Principle of Dynamics

The energy consumption of a UAV during a delivery mission is computed using a physics-based model derived from the Fundamental Principle of Dynamics (FPD). The model considers the forces acting on the drone during its vertical and horizontal flight phases, and estimates the power and energy required to execute a complete delivery task, including takeoff, cruise, delivery, and return.

3.2.1. Forces Acting on the UAV

The UAV is subjected to three main forces during flight:

- Gravitational force F_w , due to the total weight of the drone and payload:

$$F_w = (m_d + m_p) \cdot g \quad (1)$$

where m_d is the drone mass, m_p is the payload mass, and g is the gravitational acceleration.

- Aerodynamic drag force F_D , which includes both vertical and horizontal components. For a constant-speed flight, it is expressed as:

$$F_D = \frac{1}{2} \cdot \rho_{air} \cdot C_x \cdot S \cdot v^2 \quad (2)$$

where ρ_{air} is the air density, C_x the drag coefficient, S the reference surface area, and v the UAV speed.

- Thrust force F_T , required to counteract both weight and drag:

$$F_T = \sqrt{F_w^2 + F_D^2} \quad (3)$$

Power and Energy Computation

The power required by the propulsion system is calculated as:

$$P = F_T \cdot v_s. + P_{avionics} \quad (4)$$

where $P_{avionics}$ accounts for the constant power drawn by onboard electronics.

The total energy consumed during a flight segment of duration t is given by:

$$E = \frac{P \cdot t}{\eta} \quad (5)$$

where η is the efficiency of the powertrain.

3.2.2. Mission Profile and Total Energy

A delivery mission is composed of several sequential segments:

- Takeoff from the bus and vertical ascent;
- Horizontal cruise toward the delivery zone;
- Hovering and parcel drop-off;
- Return flight (cruise and descent);
- Landing back on the bus.

Each of these phases is modeled individually with its respective speed and direction. The overall profile is illustrated in Figure 2, which shows a typical mission from takeoff to return.

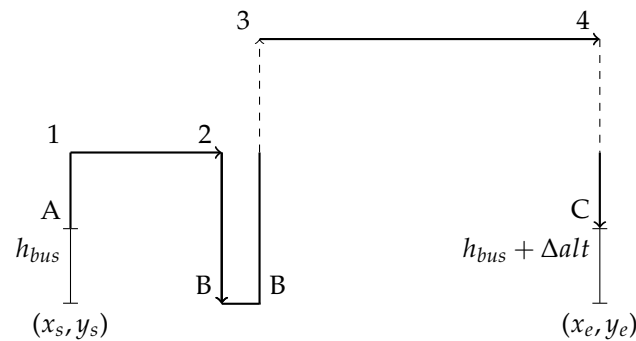


Figure 2. Delivery Profile.

The total mission energy E_{path} is computed by summing the energy consumption of each segment:

$$\begin{aligned}
 E_{path} = & \eta_{PWT} \cdot P_{A-1} \cdot \frac{d_{v,A-1}}{v_v} + \eta_{PWT} \cdot P_{1-2} \cdot \frac{d_{h,1-2}}{v_h} + \\
 & \eta_{PWT} \cdot P_{2-B} \cdot \frac{d_{v,2-B}}{v_v} + \eta_{PWT} \cdot P_{B-3} \cdot \frac{d_{v,B-3}}{v_v} + \\
 & \eta_{PWT} \cdot P_{3-4} \cdot \frac{d_{h,3-4}}{v_h} + \eta_{PWT} \cdot P_{4-C} \cdot \frac{d_{v,4-C}}{v_v}
 \end{aligned} \quad (6)$$

This formulation provides a fine-grained and realistic estimation of the energy required for each complete delivery loop.

3.3. Delivery Mission Scenarios

To assess how real-world constraints impact UAV energy consumption, we define four mission scenarios of increasing complexity. Each scenario corresponds to a modification of one or more parameters in the physical model described in Section 3.2. Table 1 summarizes the characteristics of each configuration.

- **Scenario 1—Ideal Mission:** Baseline configuration with constant speed, no external disturbances, and maximum time available for delivery. The UAV consumes energy based solely on the physical flight profile (Equation (6)), without any additional constraints or corrections.
- **Scenario 2—Delivery Time Constraint:** A delivery time is enforced based on the bus's travel time between two stops. The UAV must complete its delivery and return to the bus within this fixed time window, potentially requiring an increase in horizontal speed v_h on the return phase. This leads to increased drag and power, hence higher energy consumption.
- **Scenario 3—Aerodynamic Disturbance:** This scenario includes thrust variations due to proximity effects, such as urban obstacles and bus motion. Correction factors

derived from CFD simulations (Section 3.4) are applied to the thrust computation, altering F_T and thereby modifying P and E .

- **Scenario 4—Early Return Strategy:** In order to avoid impacting the bus schedule, the UAV is required to return before the bus reaches the next stop. This constraint reduces the available time t_h for delivery and return, requiring higher cruise speed, which further increases power and energy consumption.

Table 1. Overview of the Four Mission Scenarios.

Scenario	Constraint Applied	Parameter Affected	Effect on Energy
S1	None (ideal conditions)	–	Baseline
S2	Delivery time window	$v_h \uparrow$	Moderate increase
S3	Aerodynamic disturbance	$F_T \uparrow$	Significant increase
S4	Early bus rejoin	$t_h \downarrow, v_h \uparrow$	Maximum increase

This scenario-based structure enables a controlled and progressive evaluation of energy overconsumption factors. Each configuration is simulated with varying payloads, as detailed in Section 4.2, to assess the system’s sensitivity to operational constraints.

3.4. Aerodynamic Corrections via CFD

The physical model described in Section 3.2 assumes steady-state aerodynamic conditions during UAV flight. However, in dense urban environments, UAVs are subjected to significant airflow perturbations due to the presence of nearby obstacles, such as buildings, road infrastructure, or even the buses themselves. These perturbations affect the aerodynamic efficiency of the UAV and induce **variations in thrust**, which directly impact energy consumption.

To quantify these effects, computational fluid dynamics (CFD) simulations were conducted using ANSYS 2022. The simulations aimed to characterize the thrust variation experienced by a quadcopter-type UAV (specifically a DJI Phantom 3 (from DJI company, France)) in different flight conditions relevant to the delivery mission profile described in Figure 2.

3.4.1. CFD Setup and Configuration

The fluid domain was meshed using a tetrahedral grid composed of approximately 5.2 million elements, with refined layers near the propeller to capture boundary layer effects. The turbulence model used was the SST $k-\omega$ model, which is well-suited for capturing near-wall turbulence and wake effects. The UAV propeller was modeled in hovering conditions with steady-state airflow. Three types of environmental interactions were simulated:

1. **Proximity to a static obstacle:** The UAV hovers at a vertical distance of 0.1 m and 0.2 m above a fixed horizontal surface (e.g., building or stationary bus roof).
2. **Hovering near a moving object:** The UAV hovers 0.2 m above a surface moving horizontally at 15 m/s and 20 m/s, mimicking the aerodynamic effect of a moving bus during reboarding.
3. **Free-flight cruise at altitude:** The UAV cruises at 10 m/s in undisturbed air (used as a reference condition).

3.4.2. Thrust Variation Results

The CFD analysis yielded average thrust variation values relative to the reference condition (free-flight hover), summarized in Table 2.

Table 2. Thrust Variation Under Different Environmental Conditions

Flight Condition	Average Thrust Variation
Hovering above static surface (0.1 m)	+9.25%
Hovering above static surface (0.2 m)	+1.05%
Hovering over moving bus (15 m/s)	−10.3%
Hovering over moving bus (20 m/s)	−14.8%
Free-flight cruise (10 m/s)	−7.00%

To simplify integration into the energy model, “average correction coefficients” were derived for each flight segment of the delivery mission:

- **+5.14%** for takeoff and landing phases near static structures (used in phases A-1, 2-B, B-3, and 4-C).
- **−12.55%** for hovering over moving vehicles (used in segments 1–2 and 3–4).
- **−7.00%** for undisturbed horizontal cruise (nominal condition).

Note that during segments 1–2 and 3–4, the drone is assumed to be affected 20% of the time by moving vehicle wake (scenario 2) and 80% by open-air cruise conditions. The weighted average thrust variation for these segments is thus:

$$\Delta F_{1-2,3-4} = 0.2 \cdot (-12.55\%) + 0.8 \cdot (-7.00\%) = -8.11\% \quad (7)$$

3.4.3. Integration into Energy Model

The thrust corrections are integrated into the energy model of each scenario by modifying the thrust term F_T accordingly, which then propagates to the power and energy consumption through Equation (6). These corrections are applied in Scenarios 3 and 4 (see Table 1), which include environmental disturbances.

This CFD-based enhancement increases the accuracy of energy estimation and reflects the operational cost of flying in disturbed airspace—especially relevant in urban missions near buildings and large moving vehicles such as buses.

4. Case Study and Results

4.1. Experimental Setup

The proposed modeling framework was applied to a real-world case study based on the public transportation system (PTS) of the city of Belfort, France. The goal was to evaluate the total energy consumption of a UAV-based delivery system operating along real bus lines, under the four mission scenarios defined in Section 3.2.

4.1.1. Public Transport Network Modeling

A digital twin of the Belfort PTS was constructed using data provided by the Régie des Transports du Territoire de Belfort (RTTB). The network includes the topology of the bus lines, GPS coordinates of each stop, and scheduled inter-stop travel times.

Each bus line was modeled as a directed graph, where:

- Nodes represent bus stops and store 2D GPS coordinates and altitude (retrieved via Google Maps Elevation API).
- Edges represent road segments between stops and include the following:
 - The real travel distance (from OpenStreetMap via OSRM API);
 - Bus travel time (from RTTB data);
 - Straight-line UAV distance (computed via WGS84 ellipsoid).

This graph structure enables accurate computation of drone cruise distances and synchronization with bus movement.

4.1.2. UAV Characteristics and Parameters

The UAV considered in the simulations is a quadcopter of similar scale to the DJI Phantom series, selected for its compact footprint and suitability for urban deployment.

Table 3 summarizes the average specifications used in the model.

Table 3. UAV Parameters Used in the Simulation.

Parameter	Value
Drone mass (without payload)	1.8 kg
Payload mass range	1–10 kg
Battery capacity	5870 mAh
Average cruise speed	10 m/s
Average flight autonomy	45 min
Max operational radius (1-way)	8.2 km
Powertrain efficiency η_{PWT}	0.85
Avionics power $P_{avionics}$	5 W
Drag coefficient C_x	0.03
Air density ρ_{air}	1.225 kg/m ³

4.1.3. Simulation Protocol

For each bus line, a full delivery sequence was simulated, where the UAV performs a delivery at each stop along the line. The drone takes off from the roof of the bus, delivers a parcel near the stop, and rejoins the same or following bus.

Simulations were executed for the following conditions:

- **Mass variation:** Payloads from 1 kg to 10 kg were tested in 1 kg increments.
- **Scenarios:** All four mission scenarios (S1 to S4) were applied to each line and each payload configuration.
- **Bus lines:** Five representative lines of the Belfort network were used, with total road distances ranging from 6.78 km to 19.01 km (see Table 4).

Table 4. Total Length of Simulated Bus Lines.

Line ID	Distance (km)
Line 1	6.78
Line 2	14.74
Line 3	19.01
Line 4	11.23
Line 5	13.47

Each simulation outputs the total energy consumed by the UAV for all deliveries along the bus line. The results are analyzed in Section 4.2.

4.2. Energy Consumption Results

This section presents the results of the simulations conducted under the four mission scenarios described in Section 3.2. For each scenario, the total energy consumed by the UAV to complete deliveries along each bus line was computed for payload masses ranging from 1 kg to 10 kg.

4.2.1. Scenario 1—Ideal Flight

Figure 3 shows the total energy consumption for each bus line under ideal conditions (Scenario 1) and Table 5 shows the energy consumption range per line. The energy grows linearly with payload mass. Line 2 exhibits the steepest slope due to its longer distance.

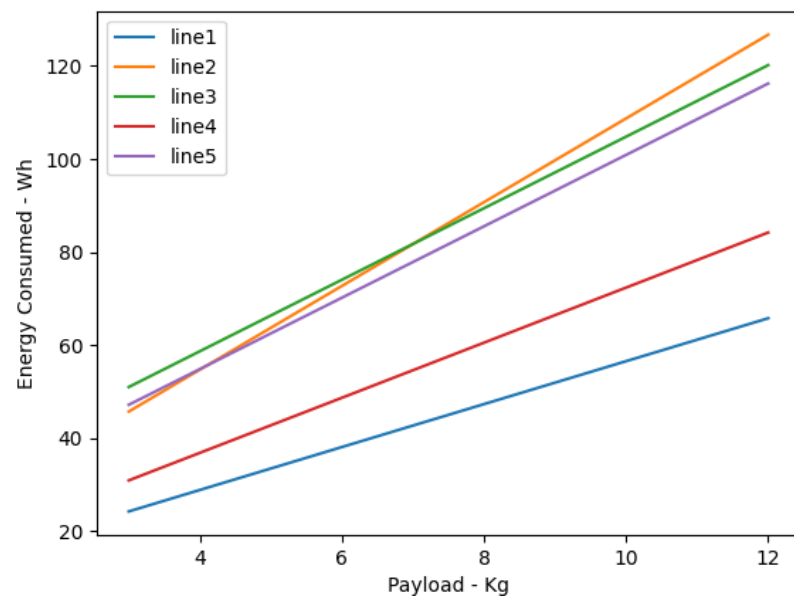


Figure 3. Energy consumption by line and payload mass in Scenario 1 (ideal mission).

Table 5. Scenario 1—Energy Consumption Range per Line.

Line	Min Energy (Wh)	Max Energy (Wh)
Line 1	24.24	65.77
Line 2	45.72	126.74
Line 3	50.99	120.18
Line 4	30.92	84.20
Line 5	47.17	116.24

4.2.2. Scenario 2—Delivery Time Constraint

In Scenario 2, the UAV must complete the delivery and return within the time window imposed by the bus's travel between stops. This leads to increased cruise speed and energy consumption (see Figure 4 and Table 6).

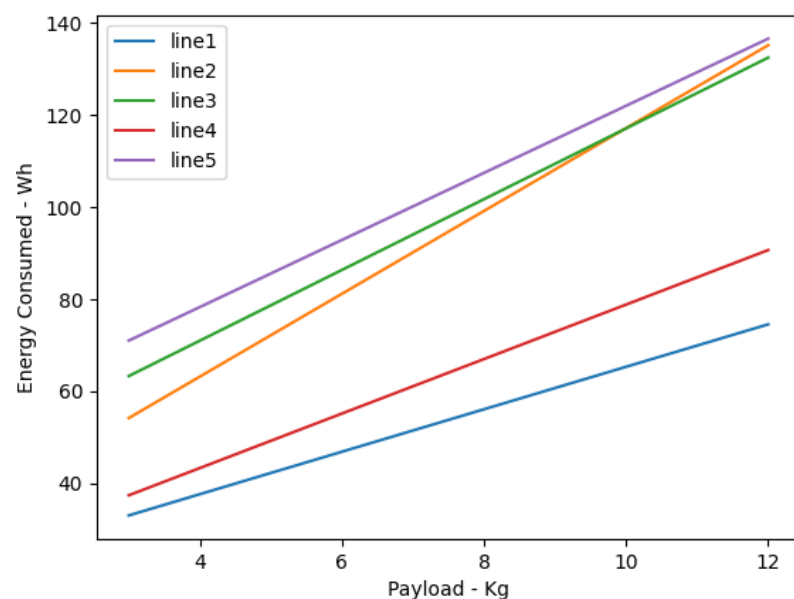


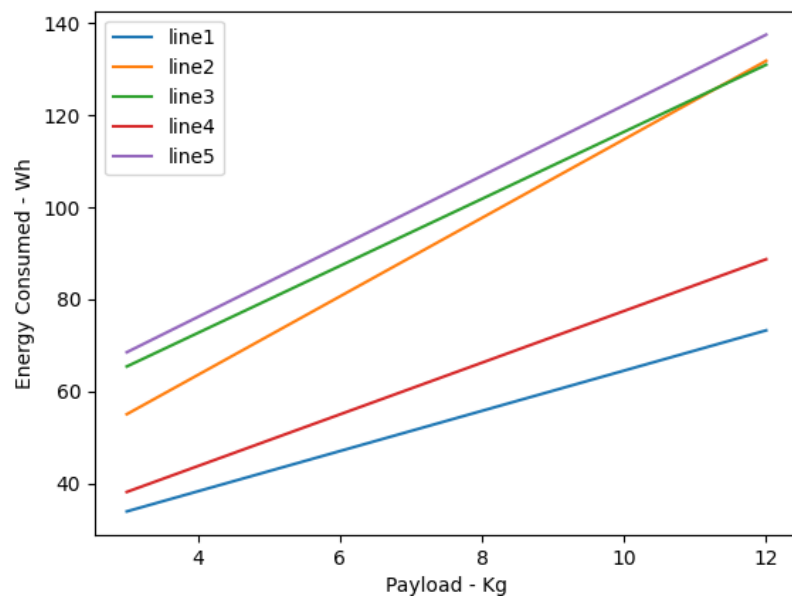
Figure 4. Energy consumption with delivery time constraint (Scenario 2).

Table 6. Scenario 2—Energy Consumption Range per Line.

Line	Min Energy (Wh)	Max Energy (Wh)
Line 1	33.03	74.56
Line 2	54.19	135.20
Line 3	63.31	132.51
Line 4	37.40	90.69
Line 5	71.02	136.63

4.2.3. Scenario 3—Aerodynamic Disturbance

In Scenario 3, the model includes thrust corrections from CFD (Section 3.4). This results in slightly different profiles, particularly in vertical phases and hovering (Figure 5 and Table 7).

**Figure 5.** Energy consumption with aerodynamic corrections (Scenario 3)**Table 7.** Scenario 3—Energy Consumption Range per Line.

Line	Min Energy (Wh)	Max Energy (Wh)
Line 1	33.87	73.25
Line 2	55.01	131.88
Line 3	65.39	131.02
Line 4	38.10	88.71
Line 5	68.49	137.55

4.2.4. Scenario 4—Early Return Strategy

Scenario 4 imposes an early return constraint to preserve the bus schedule, resulting in the highest energy consumption across all configurations (Figure 6).

These results confirm the linear increase in energy with payload mass and highlight the additive effect of each operational constraint. Scenario 4 and Table 8 shows up to **16% more energy consumption** compared to the baseline Scenario 1, justifying the need for scenario-aware planning.

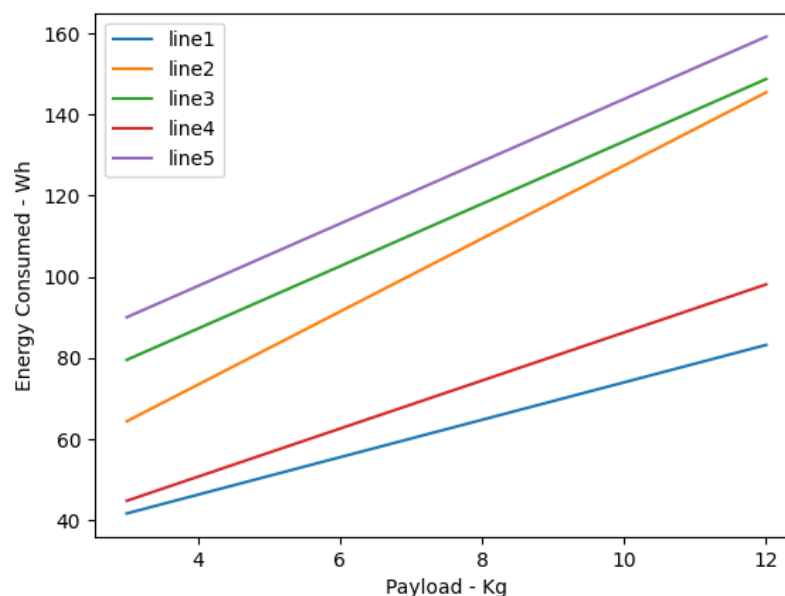


Figure 6. Energy consumption with early return constraint (Scenario 4).

Table 8. Scenario 4—Energy Consumption Range per Line.

Line	Min Energy (Wh)	Max Energy (Wh)
Line 1	41.70	83.22
Line 2	64.40	145.43
Line 3	79.53	148.73
Line 4	44.81	98.14
Line 5	90.04	159.16

4.3. Comparative Analysis

This section provides a comparative analysis across the four mission scenarios, highlighting the incremental impact of operational constraints on UAV energy consumption. The goal is to extract interpretable trends, validate the robustness of the model, and identify practical implications.

4.3.1. Scenario Impact Summary

Figure 7 compares total energy consumption across all scenarios for a 5 kg payload, representative of typical urban parcel weights. The energy consistently increases from Scenario 1 to Scenario 4, illustrating the cumulative cost of adding realistic constraints.

- Scenario 2 (delivery time constraint) results in a **10–15%** increase in energy compared to Scenario 1.
- Scenario 3 (CFD correction) adds or slightly mitigates energy depending on the delivery profile.
- Scenario 4 (early return) yields the highest overall energy consumption, with an up to **26%** increase relative to Scenario 1.

4.3.2. Relative Differences by Line

Table 9 summarizes the relative energy increases between consecutive scenarios for all bus lines, with a fixed payload of 5 kg. This comparison quantifies the marginal cost of operational constraints.

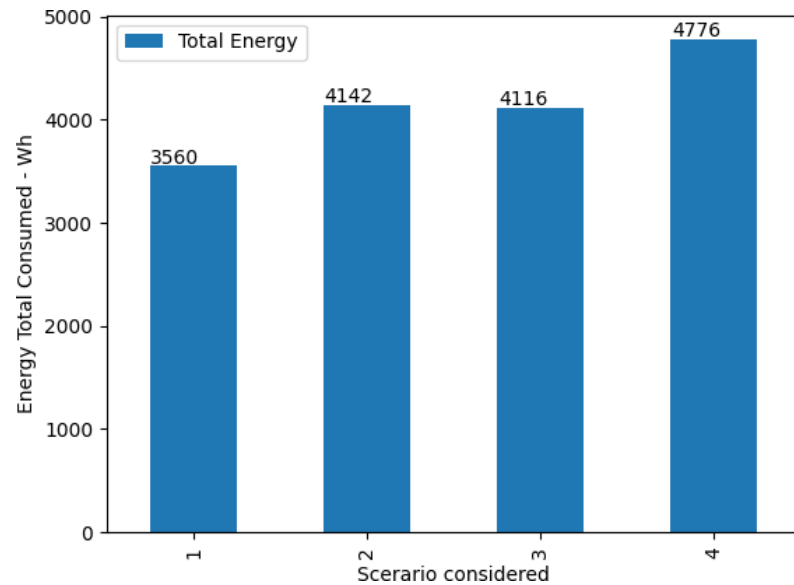


Figure 7. Comparison of energy consumption across all scenarios for a 5 kg payload.

Table 9. Relative Energy Increase Between Scenarios (5 kg payload).

Line	S2 vs. S1	S3 vs. S2	S4 vs. S3
Line 1	+12.9%	−1.8%	+10.7%
Line 2	+10.2%	−2.5%	+9.6%
Line 3	+8.9%	−1.1%	+12.2%
Line 4	+8.7%	−2.2%	+10.1%
Line 5	+11.2%	+0.7%	+12.0%

4.3.3. Mass-to-Energy Linearity

In all scenarios, the energy consumption grows linearly with payload mass. Figure 8 illustrates this behavior for Line 2. The strong correlation validates the physical consistency of the model and supports using linear approximations for battery sizing or feasibility estimation.

The slope of each line can be interpreted as a specific energy cost per kilogram, which is valuable for operational planning and comparative fleet analysis.

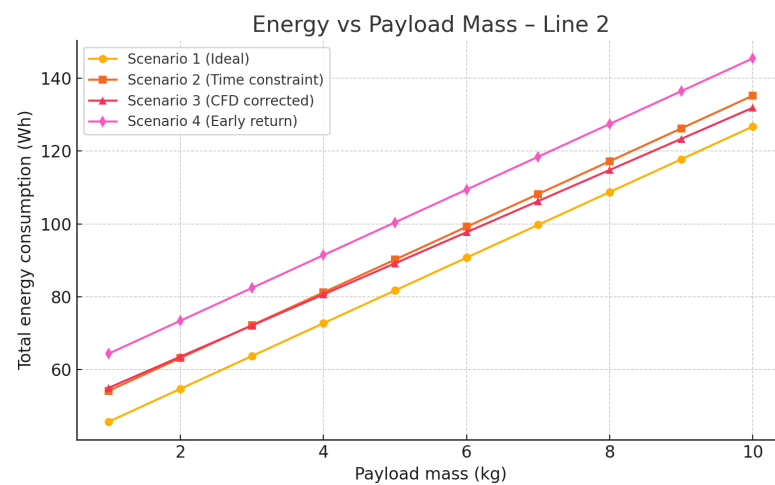


Figure 8. Linear correlation between payload mass and energy consumption (Line 2, all scenarios).

4.3.4. Operational Applications

The proposed modeling framework has several real-world applications:

- **Battery dimensioning:** Identifying the minimum required energy reserve for each delivery profile and route.
- **Mission planning:** Adjusting UAV speed and delivery timing based on bus schedules and service constraints.
- **Energy prediction:** Estimating real-time energy demand onboard to avoid infeasible missions.
- **Lifecycle planning:** Estimating battery usage and replacement needs based on urban delivery patterns.

By capturing realistic variations in power demand, this framework can be embedded in pre-flight planning tools or onboard autonomy stacks.

4.3.5. Generalization and Outlook

Although this study focuses on the Belfort public transportation system, the methodology is transferable to any structured urban area with fixed-line vehicles (e.g., metro, tram, van circuits). The modular energy model can accommodate the following:

- Different UAV types or propulsion technologies;
- Variable altitude or weather profiles;
- Multimodal vehicle couplings (bus + van, train + drone).

This generalization potential positions the model as a valuable tool for future integrated mobility systems, where drones cooperate with public infrastructure to improve delivery efficiency. These findings lay the groundwork for the conclusions presented in Section 5.

5. Conclusions and Future Work

This paper proposed a comprehensive and physically grounded modeling framework for estimating the energy consumption of UAV-based last-mile delivery missions integrated with a public transportation system. Unlike traditional models based solely on empirical correlations or geometric approximations, the proposed approach leverages a force-based energy formulation derived from the Fundamental Principle of Dynamics, corrected by aerodynamic factors obtained via CFD simulations.

The methodology was applied to a real-world case study in the city of Belfort (France), involving five bus lines, four operational scenarios, and ten payload configurations. Results confirm the model's ability to capture the linear dependency between payload mass and energy consumption, while also highlighting the significant impact of realistic constraints such as delivery time windows, bus schedule synchronization, and environmental disturbances. Notably, the most constrained scenario (early return strategy) can lead to up to 26% more energy consumption than the ideal baseline.

The modular structure of the model enables scenario-aware planning, accurate energy budgeting, and battery dimensioning for urban drone logistics. It also paves the way for integration into real-time mission feasibility tools or embedded estimation modules.

5.1. Limitations

While the model covers the physical and environmental aspects of drone energy consumption in detail, several simplifications were made:

- Wind, rain, and temperature effects were not modeled;
- Drone-bus synchronization was treated deterministically;
- Energy recovery, battery aging, and real powertrain losses were considered via average efficiency.

5.2. Future Work

Future extensions of this work will focus on the following:

- Integrating stochastic elements such as wind fields or traffic-based bus delays;
- Coupling with optimization modules for drone–bus routing or fleet sizing;
- Extending the model to multiple UAVs operating in coordinated swarms;
- Validating the framework with experimental energy data from onboard telemetry and field tests.

The framework established here offers a solid foundation for future drone–infrastructure co-design approaches, and supports the broader deployment of energy-efficient urban aerial logistics within multimodal transport systems.

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